

LESSON 3

- BOOK CHAPTER 4
(Projectile Motion)

Problem 23 (Book chapter 4):

A projectile is fired horizontally from a gun that is 45.0 m above flat ground, emerging from the gun with a speed of 250 m/s. (a) How long does the projectile remain in the air? (b) At what horizontal distance from the firing point does it strike the ground? (c) What is the magnitude of the vertical component of its velocity as it strikes the ground?

Answer: (a) We know

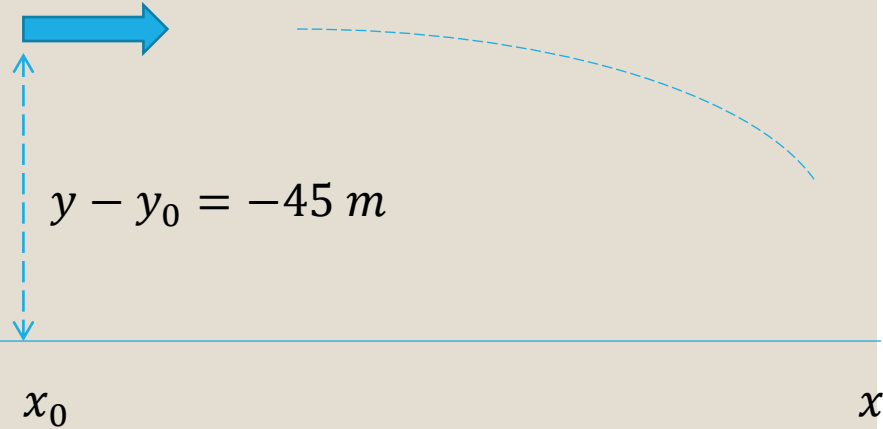
$$y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$$

$$-45 = (v_0 \sin 0^\circ) t - 4.9 t^2$$

$$-45 = 0 - 4.9 t^2$$

$$t = \sqrt{\frac{45}{4.9}} = 3.03 \text{ s}$$

$$v_0 = 250 \text{ m/s} \quad \text{and} \quad \theta_0 = 0^\circ$$



(b) We know $x - x_0 = (v_0 \cos \theta_0) t$

$$x - x_0 = (250)(\cos 0^\circ) (3.03)$$

$$x - x_0 = (250)(1)(3.03)$$

$$x - x_0 = 757.50 \text{ m}$$

(c) We know $v_y = v_0 \sin \theta_0 - gt$

$$v_y = 250(\sin 0^\circ) - (9.8)(3.03)$$

$$v_y = 0 - (9.8)(3.03)$$

$$v_y = -29.69 \text{ m/s}$$

The magnitude of v_y is 29.69 m/s

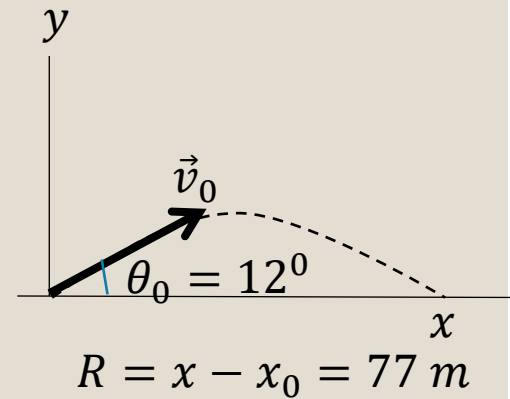
Problem 25 (Book chapter 4):

The current world-record motorcycle jump is 77.0 m, set by Jason Renie. Assume that he left the take-off ramp at 12.0° to the horizontal and that the take-off and landing heights are the same. Neglecting air drag, determine his take-off speed.

Answer:

Since the take-off and landing heights are the same, that is $y - y_0 = 0$, we can use the formula

$$R = \frac{v_0^2 \sin 2\theta_0}{g} \quad \text{or} \quad v_0 = \sqrt{\frac{Rg}{\sin 2\theta_0}}$$
$$\text{or } v_0 = \sqrt{\frac{(77)(9.8)}{\sin 24^\circ}}$$



$$v_0 = \sqrt{\frac{754.6}{0.4067}} = 43.07 \text{ m/s}$$

Problem 30 (Book chapter 4):

A soccer ball is kicked from the ground with an initial speed of 19.5 m/s at an upward angle of 45° . A player 55 m away in the direction of the kick starts running to meet the ball at that instant. What must be his average speed if he is to meet the ball just before it hits the ground?

Answer: Here, $y - y_0 = 0$

We use the following formula to find the time of flight of the ball.

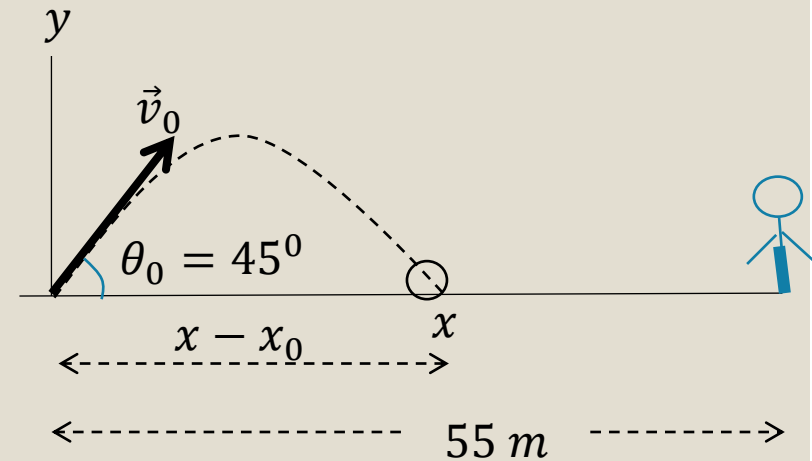
$$y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$$

$$0 = (19.5) (\sin 45^\circ) t - 4.9 t^2$$

$$(19.5)(0.707) t = 4.9 t^2$$

$$t = \frac{13.787}{4.9} = 2.81 \text{ s}$$

The player must take the time 2.81 s to meet the ball.



We need to find $x - x_0$ to obtain the distance traveled by the player.

$$x - x_0 = (v_0 \cos \theta_0) t = (19.5)(\cos 45^\circ)(2.81)$$

$$x - x_0 = 38.74 \text{ m}$$

$$\text{Average speed of the player} = \frac{\text{Distance traveled by the player}}{\text{Time taken by the player}} = \frac{55 - 38.74}{2.81} = 5.786 \text{ m/s}$$

Problem 32 (Book chapter 4):

You throw a ball toward a wall at speed 25.0 m/s and at angle 40.0° above the horizontal (as shown in the figure). The wall is distance $d = 22.0$ m from the release point of the ball. (a) How far above the release point does the ball hit the wall? What are the (b) horizontal and (c) vertical components of its velocity as it hits the wall? (d) When it hits, has it passed the highest point on its trajectory?

Answer:

(a) We know $y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$

$$y - y_0 = (25)(\sin 40^\circ) t - 4.9 t^2$$

$$y - y_0 = (25)(0.6428)t - 4.9 t^2$$

$$y - y_0 = 16.07t - 4.9 t^2$$

To find t we use the following formula,

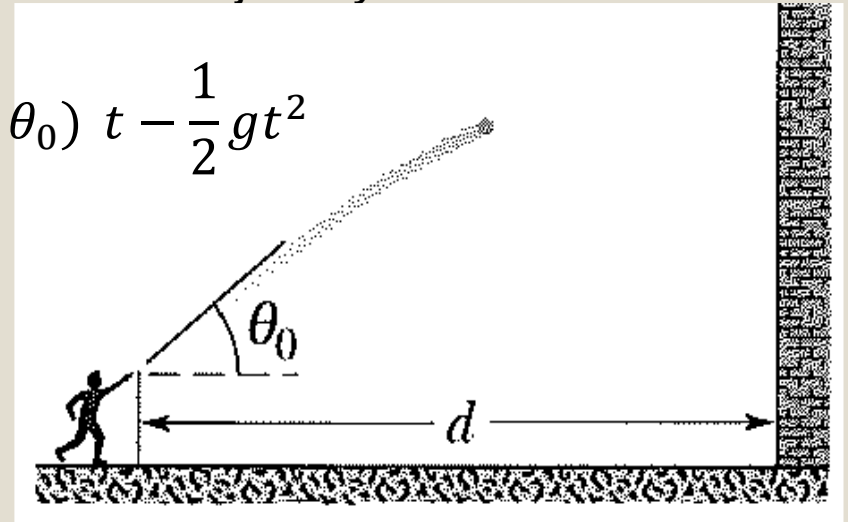
$$x - x_0 = (v_0 \cos \theta_0) t$$

$$t = \frac{x - x_0}{v_0 \cos \theta_0} = \frac{22}{25 \cos 40^\circ} = \frac{22}{(25)(0.7660)}$$

$$t = 1.149 \text{ s}$$

Therefore,

$$y - y_0 = (16.07)(1.149) - (4.9)(1.149)^2 = 18.46 - 6.469 = 11.99 \text{ m}$$



Given $v_0 = 25 \text{ m/s}$; $\theta_0 = 40^\circ$

$$x - x_0 = d = 22 \text{ m}$$

(a) $y - y_0 = ?$ (b) $v_x = ?$ and (c) $v_y = ?$

(d) Did the ball pass the highest point?

(b) We know $v_x = v_{0x} = v_0 \cos \theta_0 = 25 \cos 40^\circ = (25)(0.766) = 19.15 \text{ m/s}$

(c) We know $v_y = v_0 \sin \theta_0 - gt = 25 \sin 40^\circ - (9.8)(1.149)$

$$v_y = (25)(0.6428) - 11.26 = 4.81 \text{ m/s}$$

(d) Since v_y is positive, that is, $v_y > 0$, the ball did not reach to the highest point on hitting the wall.

Let's Practice !!

1. [Chap 4 - problem 21]: A dart is thrown horizontally with an initial speed of 10 m/s toward point P, the bull's-eye on a dart board. It hits at point Q on the rim, vertically below P, 0.19 s later. (a) What is the distance PQ? (b) How far away from the dart board is the dart?
2. [Chap 4 - problem 24]: In the 1991 World Track and Field Championships in Tokyo, Mike Powell jumped 8.95 m, breaking by a full 5 cm the 23-year long-jump record set by Bob Beamon. Assume that Powell's speed on takeoff was 9.5 m/s (about equal to that of a sprinter) and that $g = 9.80 \text{ m/s}^2$ in Tokyo. How much less was Powell's range than the maximum possible range for a particle launched at the same speed?
3. [Chap 4 - problem 29]: A projectile's launch speed is five times its speed at maximum height. Find launch angle θ_0 .
4. [Chap 4 - problem 36]: During a tennis match, a player serves the ball at 23.6 m/s, with the center of the ball leaving the racquet horizontally 2.37 m above the court surface. The net is 12 m away and 0.90 m high. When the ball reaches the net, (a) does the ball clear it and (b) what is the distance between the center of the ball and the top of the net? Suppose that, instead, the ball is served as before but now it leaves the racquet at 5.00° below the horizontal. When the ball reaches the net, (c) does the ball clear it and (d) what now is the distance between the center of the ball and the top of the net?



NEXT

BOOK CHAPTER 5

(FORCE AND MOTION-1)

Thank You